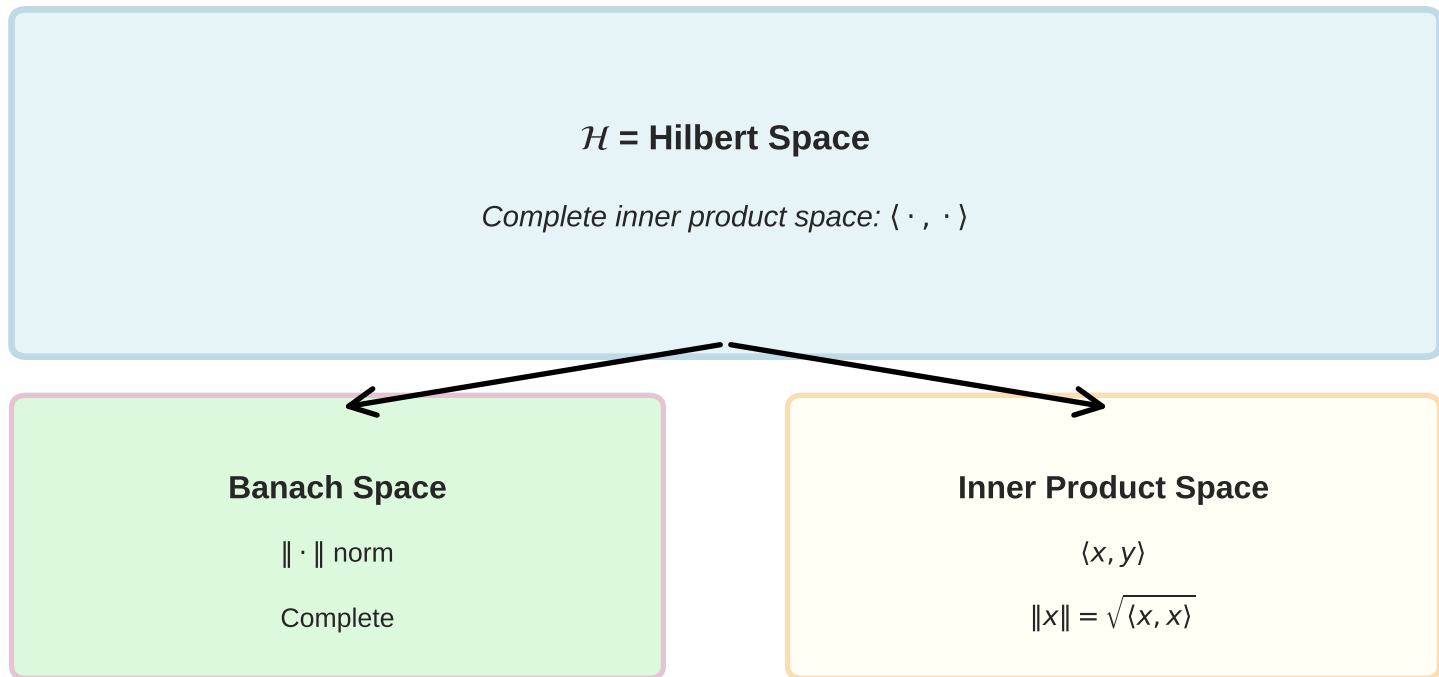


Hilbert Space Structure



Key Properties:

$\mathcal{H}$ is a Banach space with $\|x\| = \sqrt{\langle x, x \rangle}$

Parallelogram law: $\|x + y\|^2 + \|x - y\|^2 = 2(\|x\|^2 + \|y\|^2)$

Cauchy-Schwarz: $|\langle x, y \rangle| \leq \|x\| \|y\|$